Paidly — Product Architecture Strategy

Systems + Document Engine + Experience layer · with technical appendix

EXECUTIVE FRAMING

Old lens	New lens
Separate Invoices / Quotes / Payslips	**One Document Engine**, multiple **document kinds**
Features as silos	**Four core systems** + cross-cutting **Experience System**
Routes drive thinking	**Capabilities** drive architecture; routes are entry points

Part A — Core systems

1. Identity & Access

Supabase Auth, profiles, orgs, memberships, roles, RLS. **Strategy:** stabilise—harden session, org bootstrap, `RequireAuth` so other systems trust it cheaply.

2. Document Engine (commerce core)

Job: Issue documents to third parties → deliver → track → settle. **Kinds today:** Invoice, Quote, Payslip (each maps to existing entities/tables).

Shared lifecycle: Author → Compose → Render (PDF) → Deliver (email/link) → Observe (opens, reminders) → Settle (pay / accept / archive).

> **Product point:** Invest once in send, PDF, public token, status, and line-item patterns—parity across kinds is *engine work*, not three copies.

3. Relationship & Offering

Clients + catalog (`services`) + line intelligence—**input graph** into the Document Engine, not a disconnected "contacts app."

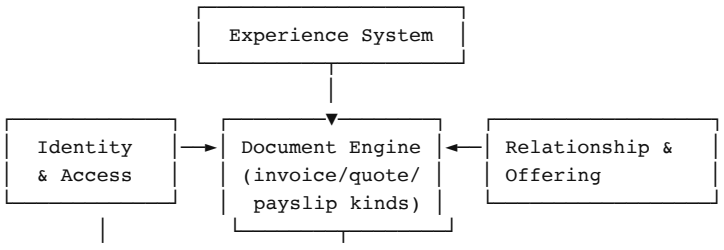
4. Revenue & Operating

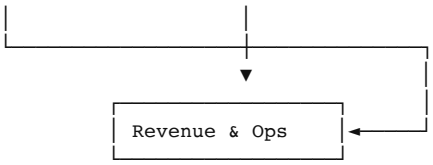
Payfast, subscriptions, crons, affiliates, admin, reporting. **Strategy:** keep thin on rules that belong inside documents (e.g. overdue logic with invoices).

5. Experience System (cross-cutting)

Pillar	Meaning
Shell	Editor pattern: work area + sticky summary/actions (EditQuote / EditClient / EditCatalogItem)
Tokens	Semantic colours—border-border, bg-card, muted-foreground
Data discipline	Query keys, invalidation, cache-first hydration where safe
A11y	Label/id parity, meaningful disabled states

SYSTEM MAP





Part B — Technical appendix

Stack: Vite + React + React Router · Tailwind + Radix · TanStack Query · Supabase · Vercel `/api/*` · Payfast · Resend · Anvil · Turnstile.

Repo: `src/pages/` (routes), `src/services/` (domain), `src/api/customClient.js` (EntityManager → Supabase), `api/` (serverless), `supabase/migrations/`.

Flow: Auth → org scope → entity CRUD + Query cache → `/api` for secrets and side effects.

BACKLOG HOOKS (STRATEGY → WORK)

1. Name Document Engine in code (`src/document-engine/` or shared module): kinds, status, send + PDF adapters.
2. Align quote / invoice / payslip on shared deliver + observe interfaces.
3. One-page Experience checklist for new money/document surfaces.
4. Identity hardening pass (session, invites, role matrix doc).